BASANT ARYAL

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Education

University of North Carolina at Pembroke Expected Dec. 2026
B.S. Computer Science, Cybersecurity Concentration *Pembroke, NC*

GPA: 4.0/4.0

Wake Technical Community College Dec. 2024
Associate in Science *Raleigh, NC*

GPA: 3.585/4.0

Technical Skills

Languages: Python, Java, JavaScript, SQL, PHP, HTML, CSS

Backend and Databases: Flask, REST APIs, JSON, MySQL

Development Tools: Git, GitHub, XAMPP, phpMyAdmin

Machine Learning and AI: TensorFlow, PyTorch, Flower, FAISS, Sentence Transformers

Additional Technologies: Streamlit, IBM Qiskit, Excel, Power BI

Experience

University of North Carolina at Pembroke Aug. 2026 – Present
Undergraduate Research Assistant, Cybersecurity *Pembroke, NC*

- Conduct research on Quantum Random Number Generation (QRNG), focusing on the security, unpredictability, and statistical properties of quantum-generated random sequences.
- Investigate quantum circuit designs using Python and IBM Qiskit to study the effects of quantum operations, noise, and measurement on randomness quality.
- Explore entropy estimation, statistical randomness testing, and NIST SP 800-22 and SP 800-90B frameworks to evaluate random number generators for cryptographic applications.
- Investigate machine learning techniques for detecting statistical bias, identifying non-random patterns, and improving the quality of quantum-generated sequences.
- Examine the implications of quantum randomness, noise, and entropy for cryptographic security and cybersecurity applications.

University of North Carolina at Pembroke May 2026 – Jul. 2026
Undergraduate Research Assistant, Computer Science and Mathematics *Pembroke, NC*

- Developed VIVA-AI, a Python and Flask application that analyzes student submissions and supports instructor-led assessment verification.
- Integrated OpenRouter-compatible LLM APIs using REST requests and JSON for submission analysis and conceptual multiple-choice question generation.
- Implemented backend workflows for submission processing, quiz generation, result handling, and instructor review, with a three-category assessment review workflow.
- Researched Canvas LMS REST API integration and co-authored a short paper accepted at IEEE TALE 2026.

Holiday Inn Express & Suites Dec. 2025 – Aug. 2026
Data Analyst Assistant (Intern) *Southern Pines, NC*

- Cleaned and reviewed hotel data using Python, SQL, and Excel to support reporting and analysis.
- Analyzed occupancy, ADR, RevPAR, revenue, and booking trends; prepared Excel and Power BI reports to support operational decisions.

Publications and Accepted Papers

Selvarajah Mohanarajah, **Basant Aryal**, and Prabina Shrestha. "VIVA AI: A Privacy-Preserving AI-Assisted Assessment Verification Framework for Learning Management Systems." *2026 IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE)*. **Accepted short paper**, 2026.

Projects and Research

Cyber Evidence — Cybersecurity Multi-Agent RAG Python, FAISS, Streamlit

- Built a cybersecurity question-answering application with Query, Retrieval, Answer, and Verification agents using approved NIST documents.
- Implemented PDF processing, document chunking, and semantic search using Sentence Transformers and FAISS, returning evidence-backed answers with source citations.
- Developed a Streamlit interface and safeguards for ambiguous, unsupported, and out-of-scope questions; passed 59 automated tests.

Schedule Buddy — Professor Scheduling System Python, Flask, MySQL

- Developed a scheduling application with approval workflows, room allocation, and conflict detection.
- Built a responsive frontend using HTML, CSS, and JavaScript, with a Flask backend and MySQL database.

UNCP ACM Chapter Website PHP, MySQL, HTML/CSS

- Built a database-backed chapter website with user authentication and a member dashboard.
- Designed responsive pages for events, member information, and organizational announcements.

Network Traffic Anomaly Detection — Capstone Project Python, Machine Learning

- Developed an Isolation Forest pipeline for IoT network anomaly detection, including feature analysis and investigation of benign false positives.
- Compared Isolation Forest predictions with a teammate's autoencoder on 30,082 held-out records, verifying record alignment using evaluation IDs.
- Evaluated attack detection, false-positive rates, and combined-model strategies; project and research paper preparation are ongoing.

PlantVillage Classification and Federated Learning Python, TensorFlow, Flower

- Build and compare CNN, ResNet50, MobileNetV2, and convolutional autoencoder models using 20,638 PlantVillage images across 15 classes.
- Evaluate centralized and federated learning approaches for collaborative plant disease classification without sharing raw images between clients.
- Prepare comparative results and methodology for an ongoing research paper; manuscript in preparation.

Professional Development

Cybersecurity First Responders Training Oct. 13–16, 2026

32-hour training program.